

Week 1: The Evolution of Data

Lesson 1: The Evolution of Data

Evolution of Data

Page 1: Introduction to the Evolution of Data

Welcome to the first module of *Data in the World*, named **The Evolution of Data**. Our story begins **30,000 years ago** with the earliest forms of data: **cave paintings**.

These were not just pretty pictures — they were a crucial way for early humans to **record and share vital information** such as:

- Successful hunts
- Important rituals

These paintings acted as the very first **data points**, used to capture and share knowledge within the community.

Page 2: Symbols and Early Writing Systems

As human civilization advanced, so did our methods of recording information. We moved from simple pictures to more complex **symbols and alphabets**, like **Egyptian hieroglyphs**.

This advancement allowed:

- More detailed storytelling
- Communication of nuanced information

It shows how data has always been about making information **smarter and more accessible**, even in primitive times.

Page 3: Papyrus, Parchment & Preservation of Data

Humanity took big leaps with improvements in **materials and technology**.

Ancient civilizations developed:

- **Papyrus scrolls** in Egypt
- **Parchment** in Rome

These materials were revolutionary because they:

- Allowed data to be stored longer

- Preserved knowledge across generations
 - Prevented information from being lost
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Page 4: The Printing Press – A Game Changer

The real breakthrough came with the **printing press** in the **1400s**.

Before its invention:

- Books were hand-copied
- They were rare and expensive

After the printing press:

- Knowledge became widely accessible
- Books were available to the general population
- Spread of information accelerated dramatically

This democratized knowledge for the first time in human history.

Page 5: Typewriters, Files & Data Portability

In more recent history, tools like:

- Typewriters
- Filing systems

became common.

They helped:

- Record facts
- Maintain personal and business records
- Streamline documentation

With these inventions, knowledge became **portable**, moving into the hands of many, instead of staying confined to scrolls or walls.

Page 6: The Digital Revolution – Computers

The 20th century brought a monumental leap: **computers**.

This revolution changed everything:

- Information could be typed instantly
- Data could be copied easily
- Files could be stored digitally

We moved from dusty physical files to **digital bits — zeros and ones**, the foundation of all digital information.

Digitization made data:

- Efficient
- Easier to manage
- Easy to replicate

Page 7: Data Everywhere – Phones, Watches & Cloud

Fast forward to today: **data is everywhere**.

It is no longer confined to filing cabinets or desktop computers. Now it lives:

- In our pockets (phones)
- On our wrists (smartwatches)
- Across global networks (the cloud)

Data has gone from being **static and isolated** to a **dynamic, ever-present companion**.

Page 8: The Internet – Connecting the World

The true explosion of data began with the public release of the **Internet / World Wide Web**.

This was a defining moment because it:

- Connected people globally
- Changed how we interact and share data

New communication methods spread rapidly:

- Emails
- Websites
- Online chats

For the first time, information traveled **instantly across continents**, far faster than traditional mail.

This speed and connectivity laid the foundation for the huge data generation we see today.

Page 9: Smartphones, Digital Trails & IoT

Once the internet became widespread, the world truly started “talking” — generating unimaginable amounts of data.

Every online action creates a **digital trail**, for example:

- Browsing social media
- Watching videos
- Online shopping
- Sending emails

Smartphones accelerated data generation:

- High-resolution photos
- GPS location data
- Voice commands

Then came the Internet of Things (IoT):

Sensors were added to:

- Walls
- Buildings
- Kitchens
- Forests
- Glaciers

These sensors generate continuous data about the world around us.

Staggering figures:

- 6,000+ tweets **every second**
- 100,000+ Google searches **every second**

These numbers continue to rise every year.

Page 10: Big Data → Big Opportunities

We now understand the huge volume of data we generate. But how does this overwhelming amount become useful?

Companies use **data analytics** to:

- Improve their services
- Predict trends
- Personalize user experiences

Examples:

1. Google Maps

- Analyzes real-time location data from millions of users
- Predicts traffic
- Suggests fastest routes

2. Netflix

- Analyzes your viewing history, ratings, and preferences
- Recommends movies and shows you are likely to enjoy

These examples show how effectively analyzed data leads to:

- Smarter services
- Useful predictions
- Better user experiences

Data, when used well, becomes an **immense opportunity**.
